Leading asymptotic of the standard integral for polynomial data: only $\xi(0)$ and $\eta(0)$ survive

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Abstract

The capstone of the one-dimensional programme: for polynomial data $\xi(u) = \xi_0 + \sum_{j<D} c_j u^{j+1}$ and $\eta(u) = \sum_{i \le E} e_i u^i$ with $\eta(0) \neq 0$,

$$\int_0^b u^h \eta(u)\, e^{-\beta n u^{2k} + \beta\sqrt{n} u^k \xi(u)}\, du \;\sim\; \eta(0)\, \tfrac{1}{2k}\, S_{(h+1)/(2k)}(\xi(0))\, n^{-(h+1)/(2k)} \qquad (n \to \infty),$$

a genuine `Asymptotics.IsEquivalent`. This is the leading term of `cor:standardintegralexp` in one dimension for general (polynomial) data: the leading exponent is the first element of $\Lambda(h,k)$ and only the chart-origin values $\xi(0), \eta(0)$ enter the coefficient. En route, the polynomial-ξ two-term bound of unit 32 is packaged as $Z - \frac{1}{2k} n^{-(h+1)/(2k)} S(\xi_0) = O(n^{-(h+2)/(2k)})$. Zero `sorry`/`axiom`.

1 Setting

Units 27–32 establish the expansion for $\eta \equiv 1$ and for monomial or polynomial ξ. The paper's standard integral carries a weight η (Jacobian and observable factors); by linearity a polynomial η redistributes the problem over shifted powers u^{h+i}, each contributing at order $n^{-(h+i+1)/(2k)}$. So the $i = 0$ term dominates and the rest is $O(n^{-(h+2)/(2k)})$, and the polynomial-ξ correction is of the same order.

2 The things formalised

```
theorem isLittleO_rpow_neg_rpow_neg {a b :  } (hab : b < a) :
    (fun n :   => n ^ (-a)) =o[atTop] fun n :   => n ^ (-b)
theorem isBigO_rpow_neg_rpow_neg {a b :  } (hab : b   a) :
    (fun n :   => n ^ (-a)) =O[atTop] fun n :   => n ^ (-b)

theorem standardIntegral1D_polynomial_leading_isBigO ... :
    (fun n => Z_chart( ) n - (1/(2k)) n^(-(h+1)/(2k)) S_{(h+1)/(2k)}( ))
      =O[atTop] fun n => n ^ (-((h: )+2)/(2k))

theorem standardIntegral1D_weighted_leading_isEquivalent ... (he0 : e 0   0) :
    (fun n =>   u in Ioc 0 b, u^h * (  i   range (E+1), e i * u^i) * exp(... (u)))
      ~[atTop] fun n => e 0 * ((1/(2k)) * S_{(h+1)/(2k)}( )) * n^(-(h+1)/(2k))
```

3 Proof strategy

The power helpers are `isLittleO_of_tendsto'` with `tendsto_rpow_neg_atTop` (ratio $n^{-(a-b)} \to 0$) and `IsBigO.of_bound 1` with `rpow_le_rpow_of_exponent_le`. The polynomial-ξ big-O is unit 29's packaging applied to unit 32: the D first-order terms and the remainder are each dominated by $n^{-(h+2)/(2k)}$ for $n \geq 1$ (exponent comparison), the tails by the exponential scales. The capstone splits η by `Finset.sum_range_succ'` into e_0 and the rest, rewrites the weighted integral by linearity (`Finset.mul_sum`, `integral_finsetSum`), and shows the difference from $e_0 A_0 n^{-\mu_1}$ is $O(n^{-\nu})$ with $\nu = (h+2)/(2k)$: the e_0 part by the big-O at h, each $i \geq 1$ part by the big-O at $h+i$ plus its own leading term, all pulled down to $n^{-\nu}$ by the power helper and summed with `IsBigO.sum`. Then $n^{-\nu} = o(n^{-\mu_1})$ and `IsLittleO.const_mul_right` with the nonzero coefficient $e_0 A_0$ give `IsEquivalent`.

4 Roadblocks and resolutions

- `gcongr` on $\frac{h+2}{2k} \leq \frac{h+2+1}{2k}$ paired the summands as 2 against 1 and left '$2 \leq 1$'; for exponent comparisons use `div_le_div_iff_of_pos_right` then `linarith`. Likewise `gcongr` on $n^{-X} \leq n^{-\mu}$ reduces to the *exponent* inequality, not the power inequality one has in hand — pass explicit `mul_le_mul` chains.
- A `calc` step with '_ + _' placeholders in the middle expression was unified against the wrong subterms; spell the intermediate expression out (or name the bound as a separate `have`).
- `(IsBigO.add ?_ ?_).congr_left …` left a typeclass problem stuck because the two functions were undetermined; name both big-O facts first. `IsBigO.sum` produces a Pi-type `Finset.sum` of functions, converted to a pointwise sum by `Finset.sum_apply` inside `congr_left`.
- The final `congr'` with `EventuallyEq.rfl` also got stuck; `IsLittleO.congr_left` with the linearity identity is the robust route.

5 What was Mathlib, what was new

Mathlib: `tendsto_rpow_neg_atTop`, `isLittleO_of_tendsto'`, `rpow_le_rpow_of_exponent_le`, `IsBigO.sum`, `IsBigO.const_mul_left`, `IsLittleO.const_mul_right`, `Finset.sum_range_succ'`, `Finset.sum_apply`. New: the two power-comparison helpers and the leading asymptotic for polynomial (ξ, η).

6 Lessons

`gcongr` is excellent when the shapes on both sides match term-for-term and treacherous otherwise — with sums of different lengths or when a monotone function (here `rpow` in the exponent) rewrites the goal into a form different from the hypothesis one holds. In asymptotic bookkeeping, name every big-O fact with its function written out before combining.

7 Outcome

The one-dimensional analytic content of grammar §4.2 is now formalised for polynomial data: exact expansions, explicit remainders, and asymptotic statements at leading, second and all orders (units

22–33). The remaining gaps are structural rather than analytic: general analytic ξ, η (limits of the polynomial statements), the multivariate tree with its $\log n$ powers, and §4.3's stochastic layer.